

WEEK 1



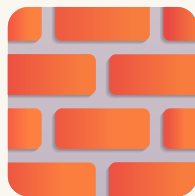
Agent Movement

Tell the robot where to walk

에이전트(로봇)에게 길 알려주기

BIG PICTURE

8 weeks → a whole village 🏠



It all starts with one step forward.

한 걸음부터 마을까지

HOW TO COUNT

Count the axis — one block, one number



1 Stand on the start block — that spot is $(0, 0, 0)$.

2 $x \rightarrow$ walk **right**. Add **+1** for every block.

3 $y \rightarrow$ jump **up**. Add **+1** for every block.

4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

What is the agent ?

The agent is your **helper robot**. You give it commands, it follows them **in order**.

```
move forward  
move forward  
move forward
```

3 commands → 3 steps. The agent does them top to bottom.

명령은 위에서 아래로 차례대로 실행돼요

CONCEPT

Four movement commands

move forward
step ahead

turn right
spin in place

move up by 3
climb 3 blocks

move down by 1
drop 1 block

앞으로 · 돌기 · 위로 · 아래로

CONCEPT

Turning ≠ moving



turn right

faces a new way
(stays put)



move forward

walks one block
(changes spot)

돌기는 방향만, 이동은 위치를 바꿔요

CODE

Blocks: walk an L-shape

```
# go forward, turn, go forward  
move forward  
move forward  
turn right  
move forward  
move forward
```

The agent traces an **L** on the ground.

CODE

The same thing in Python

```
agent.move(FORWARD, 2)
agent.turn(RIGHT_TURN)
agent.move(FORWARD, 2)
```

FORWARD = direction. 2 = how many blocks.

블록 코드와 똑같은 동작이에요

PREDICT 🤔

What shape is traced?

Predict first — then click reveal

```
move forward  
turn right  
move forward  
turn right  
move forward  
turn right  
move forward  
turn right
```

Reveal output 🔍 reveal

VOCAB

New words today

agent your helper robot that follows commands

forward the direction the agent is facing

turn change facing without moving spot

command one instruction you give the agent

에이전트

앞으로

돌다

명령



DEBUG

What's wrong with this code?

Goal: walk **5 steps forward**.

buggy

```
move forward  
move forward  
move forward
```

Think first — then click next.



DEBUG

The bug

Bug: only 3 commands, but the agent needs 5 steps. Count the commands — each `move forward` = one block.

fixed

```
move forward
```

```
move forward
```

```
move forward
```

```
move forward
```

```
move forward
```

YOUR TURN!

Build an L-path

✓ Open your homework world + Code Builder

✓ Use `move forward` to walk a few steps

✓ Add one `turn right`

✓ Walk forward again — that's the L

✓ Send a photo of where the agent stopped



QUIZ

QUESTION 1 OF 3

How far does **move forward** move the agent?

To the edge of the world.

It only turns, never moves.

Three blocks ahead.

Exactly one block ahead.



QUIZ

QUESTION 2 OF 3

What does `turn right` do?

Changes which way the agent faces, without moving.

Sends the agent home.

Places a block.

Walks one block to the right.



QUIZ

QUESTION 3 OF 3

In what order does the agent run your commands?

All at once.

Top to bottom, one at a time.

Bottom to top.

In a random order.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Agent on the move!

Next: placing blocks.

다음 주: 블록 놓기